pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Visualization

WEEK 3, DAY 1

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The desk lead's note: show me the shape

Week 2 gave me numbers: a mean spread of 185.75 bp, 128 bonds investment grade, the book by sector. Numbers hide the shape. Show me how far the spread tail runs, how spreads step up by rating, where the money sits, and what moves together. Title and label every chart; I will forward them.

11.1 Week 2's Book and the Credit Order in One Cell

```
import matplotlib.pyplot as plt
import seaborn as sns
book = pd.merge(holdings, ratings, on='rating')
bucket_order = ratings.sort_values('score')['bucket'].unique().tolist()
print(book.shape)
print(bucket_order)
(200, 22)
['AAA', 'AA', 'A', 'BBB', 'BB', 'B', 'CCC']
```

A text sort would put A before AA, as pandas Part 2 showed.

One table and one credit order, written once: every chart today draws from them.

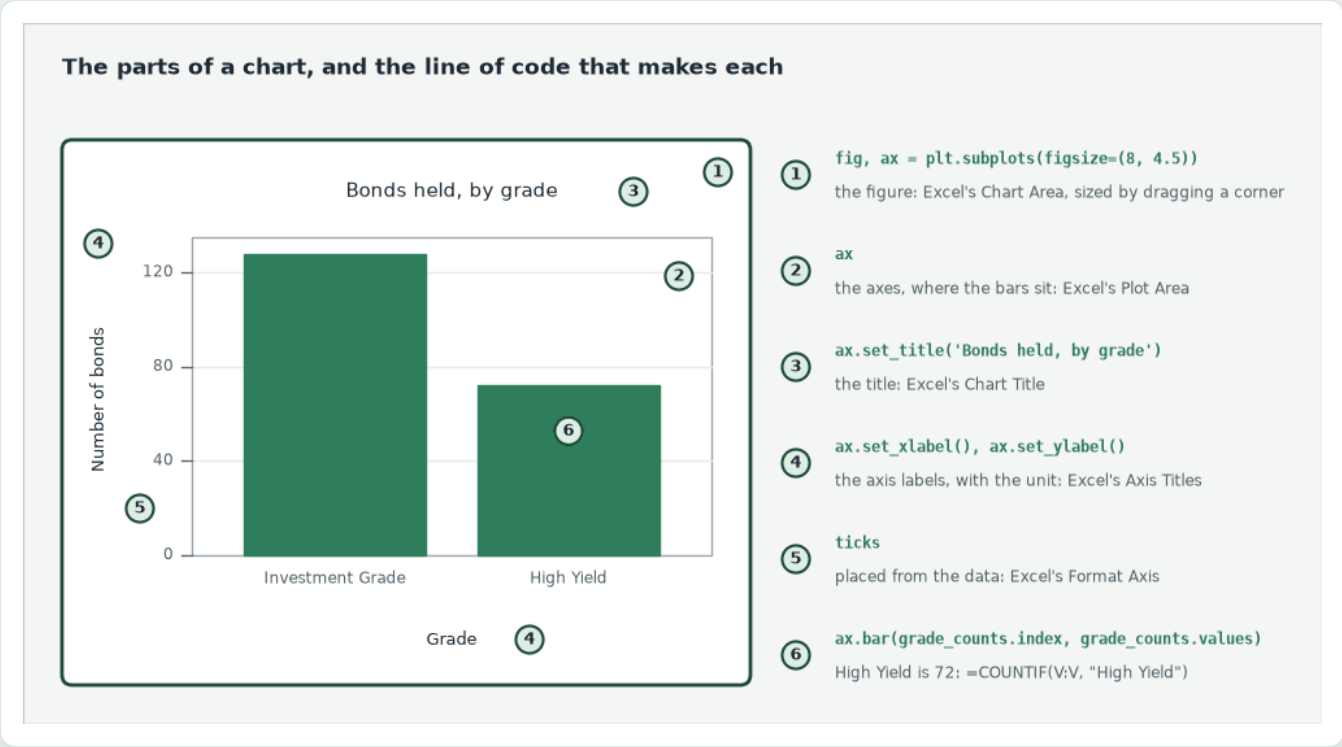
The issuers are invented. Every chart describes the data; none is a view.

Two libraries, one table: matplotlib draws, seaborn draws from the table in one line.

Excel side by side

With the scale's score in column B and its bucket in column C, "=UNIQUE(SORTBY(C2:C19, B2:B19))" in Excel 2021 and Microsoft 365.

11.2 The parts of a chart, and the line that makes each



Make, draw, label, show: every chart today is built in that order.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
plt.show()
```

Nothing is drawn, so both axes run from 0 to 1.

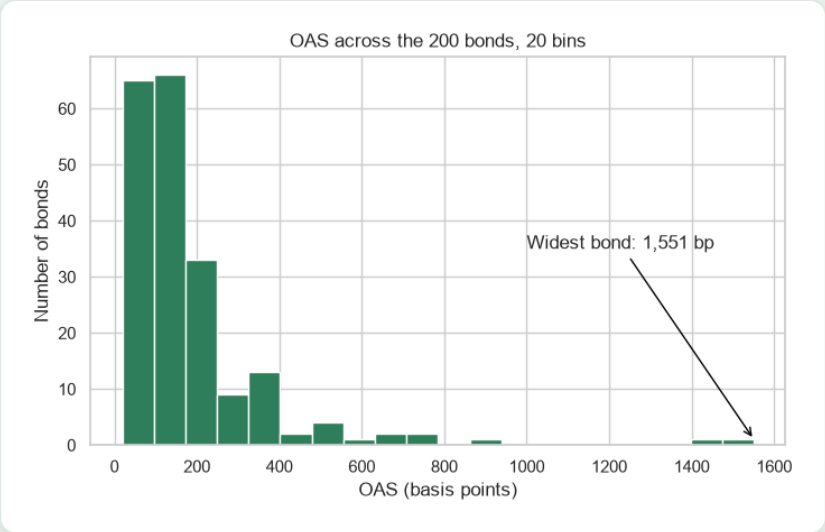
```
fig, ax = plt.subplots(figsize=(8, 4.5))
ax.bar(grade_counts.index, grade_counts.values)
ax.set_tittle('Bonds held, by grade')
ax.set_xlabel('Grade')
plt.show()
AttributeError: 'Axes' object has no attribute 'set_tittle'
```

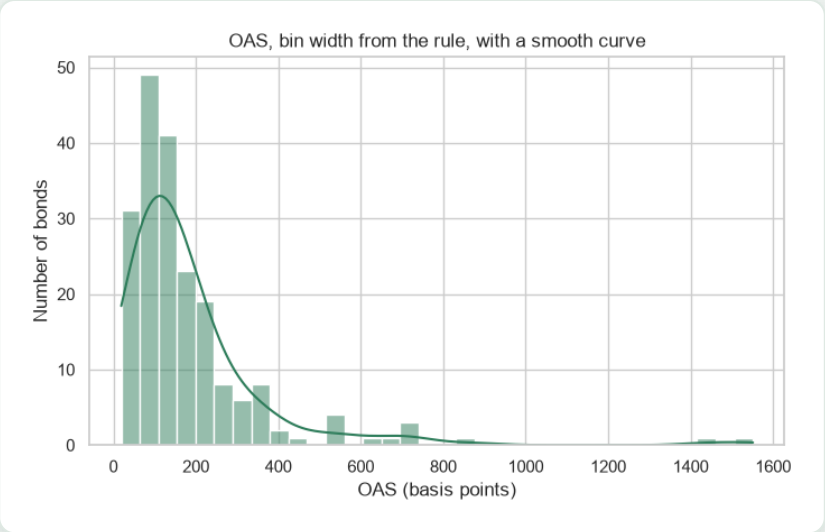
The bars were drawn before Python reached the typo, and the labels never ran: correct and unfinished, since a number with no label is a column with no header.

Excel side by side

Under each line of code, the picture names the part in Excel.

11.3 The spreads in 20 bins, and a width from a rule





Read up from the x axis: the bonds crowd the left, and a thin tail runs out to 1,551. A long tail on the right makes the column skewed right.

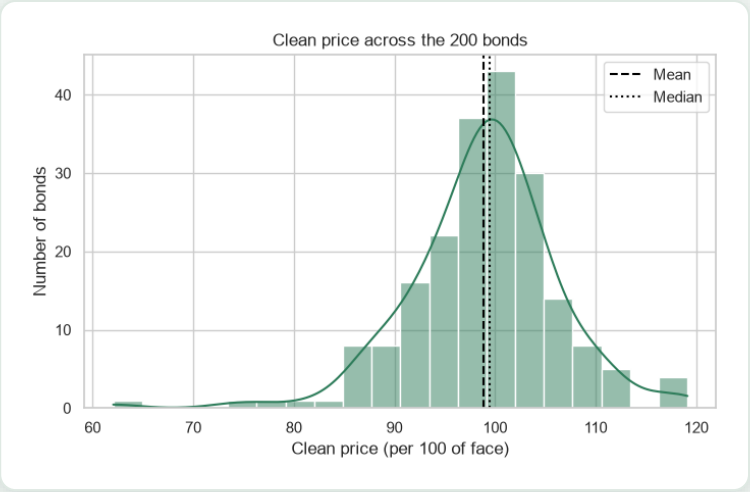
The curve is a kernel density estimate (KDE), the bars with the steps rubbed out; it peaks near 120 bp. Read counts off the bars and shape off the curve.

```
q1 = book['oas'].quantile(0.25)
q3 = book['oas'].quantile(0.75)
iqr = q3 - q1
rule_width = 2 * iqr / len(book) ** (1 / 3)
print('First quartile:', q1, '  third quartile:', q3, '  IQR:', iqr)
print('Width from the rule:', round(rule_width, 2), 'bp')
First quartile: 81.75   third quartile: 213.0   IQR: 131.25
Width from the rule: 44.89 bp
```

The IQR, the interquartile range, is the third minus the first: the width of the middle half. A rule of thumb sets a bin width of 2 x IQR over the cube root of the count.

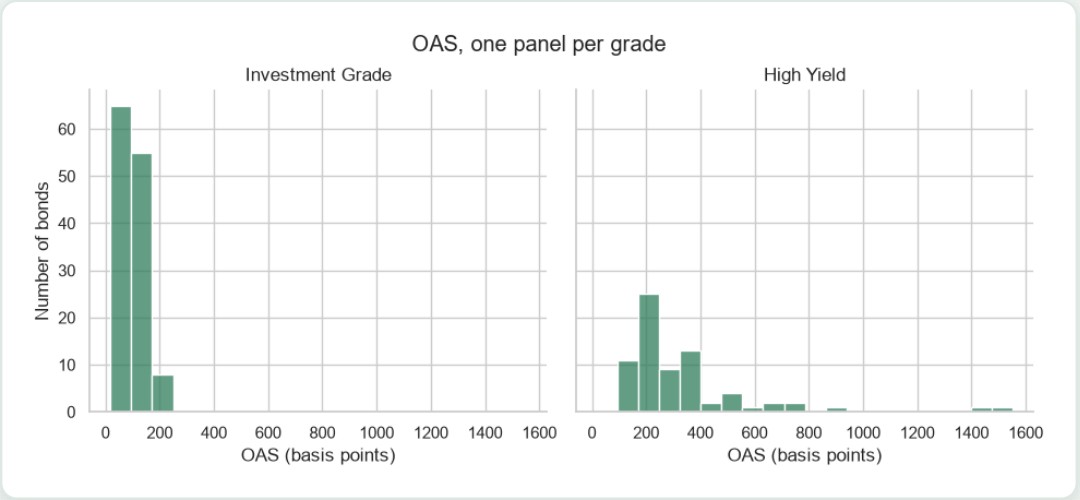
2 x 131.25 / 5.85 is 44.89 bp, and 1,532 bp of range over 44.89 is about 34 bars: a start, then look.

Skewed left, and one panel per grade



The tail runs left, to 62.05, and pulls the dashed mean below the dotted median: skewed left.

A histogram shows one column's shape; to compare groups, five numbers each are easier.



These are small multiples: Investment Grade is a compact hill from 19 to 204 bp, and High Yield starts at 125 and carries the whole tail. Seven buckets would be too many panels.

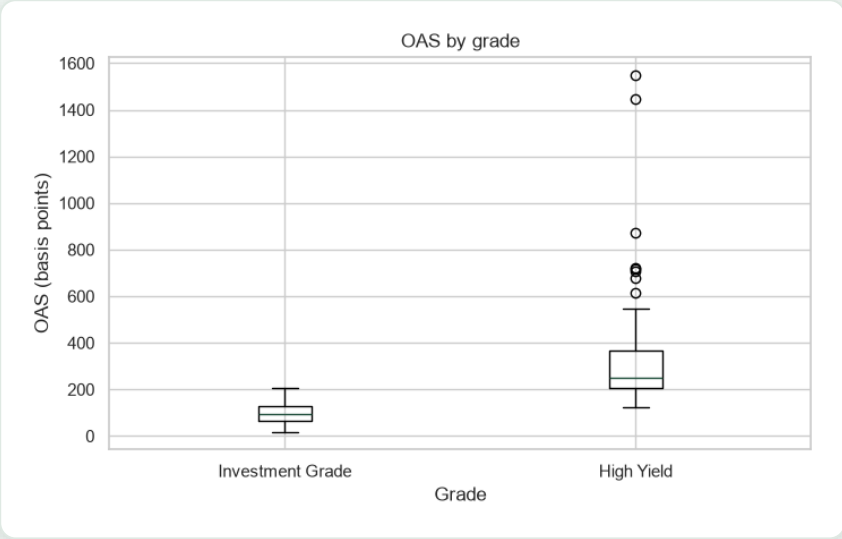
Excel side by side. The OAS is column M.

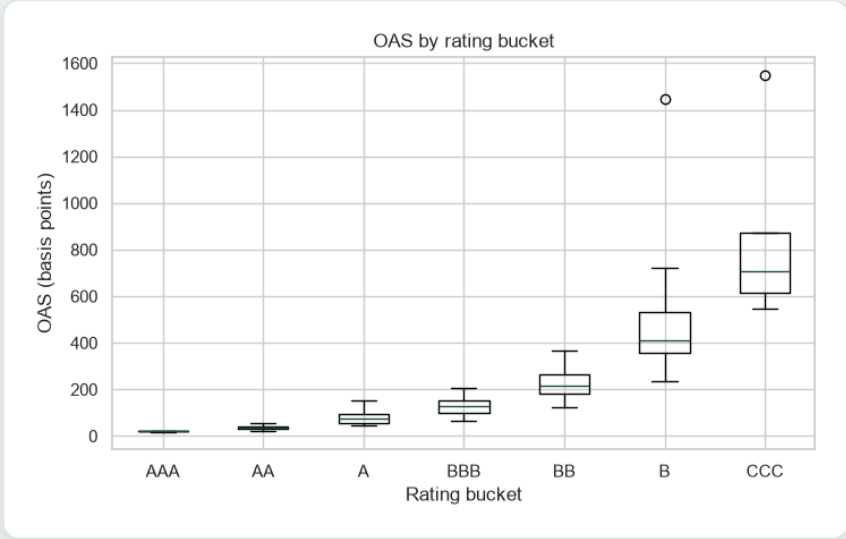
Job	Python	Excel
Histogram	`ax.hist(book['oas'], bins=20)`	Insert, Statistic Chart, Histogram
A note with an arrow	`ax.annotate(...)`	Insert, Shapes: an arrow and a Text Box

A bond on a bin edge lands in the left bar in Excel, as with Combining Part 2's "pd.cut", and in the right one in matplotlib.

11.4 Five numbers in a box: by grade, then by bucket

A box plot draws five numbers: lowest, first quartile, median, third quartile, highest. The box spans the quartiles with a line at the median; a whisker reaches toward a limit 1.5 x IQR beyond the box, and bonds past it are drawn alone.





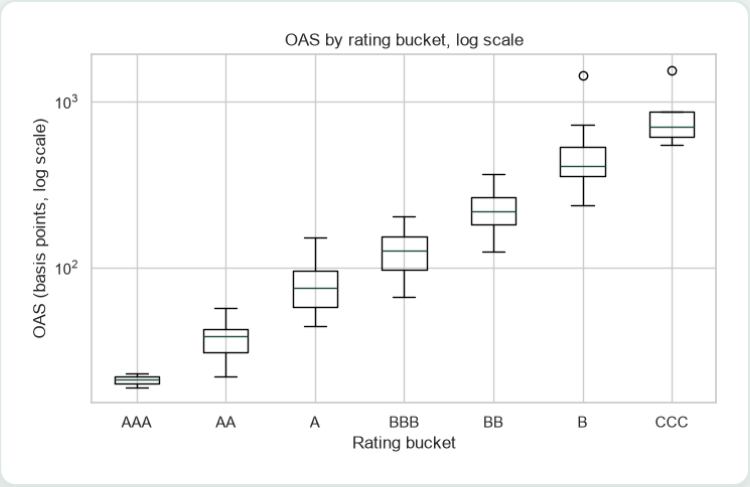
The boxes sit apart while the whiskers overlap.

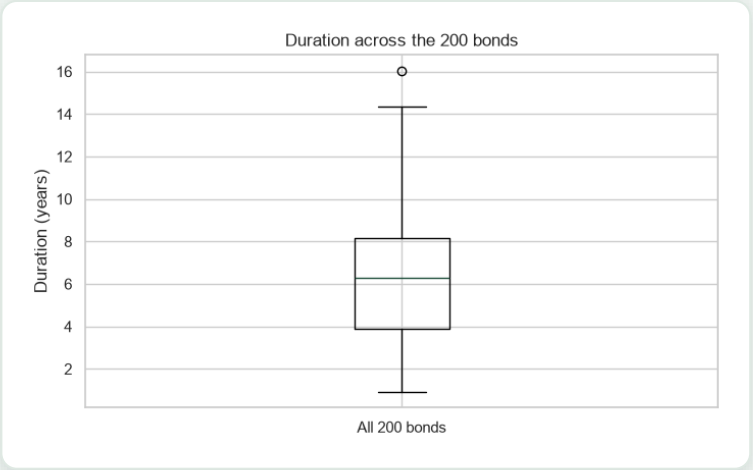
The medians step up from 21 bp at AAA to 707 at CCC, and the boxes grow taller. Bonds at 1,446 and 1,551 stretch the axis and flatten the boxes on the left.

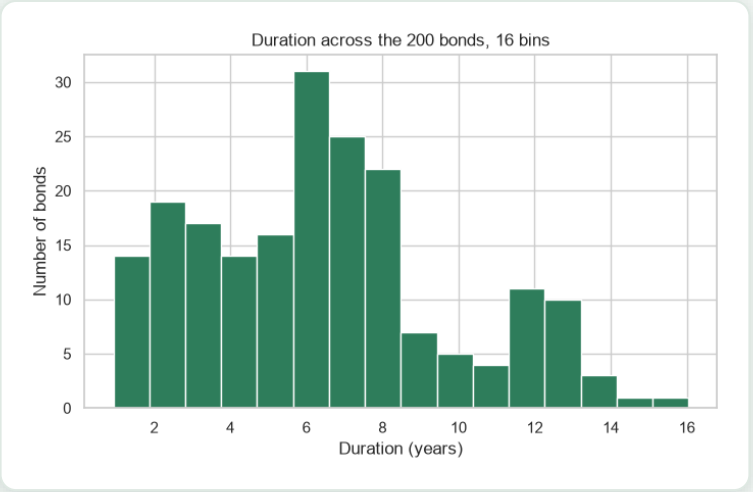
```
print(book['bucket'].value_counts().loc[bucket_order].to_dict())
{'AAA': 2, 'AA': 8, 'A': 56, 'BBB': 62, 'BB': 49, 'B': 18, 'CCC': 5}
```

Count before reading a box: AAA's is 2 bonds and CCC's 5, drawn like BBB's 62.

A log scale, and a tidy box over two humps







10 to 100 is as far as 100 to 1,000: equal distances are equal ratios, so say so in the label.

The box is tidy, with one flier at 16.04 years; the histogram shows a second hump, which a box cannot.

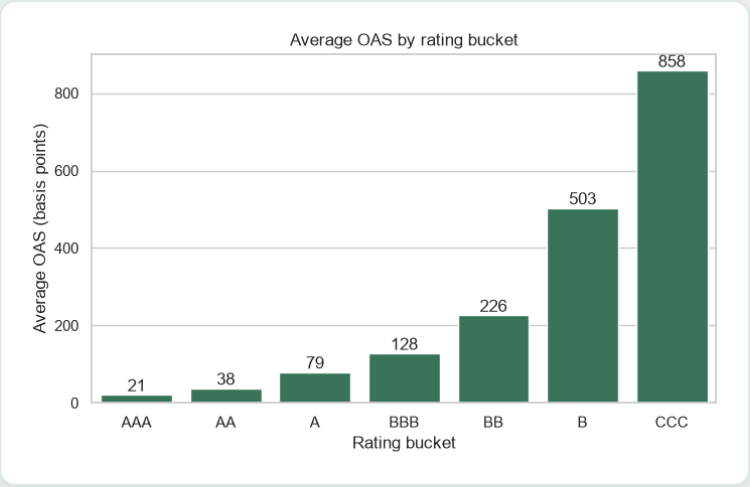
Excel side by side

Job	Python	Excel
Box plot	`ax.boxplot(book['oas'])`	Insert, Statistic Chart, Box and Whisker
Quartiles	`quantile(0.25), 81.75`	=QUARTILE.INC(M2:M201, 1), and 3 for 213
Log scale	`ax.set_yscale('log')`	Format Axis, Logarithmic scale

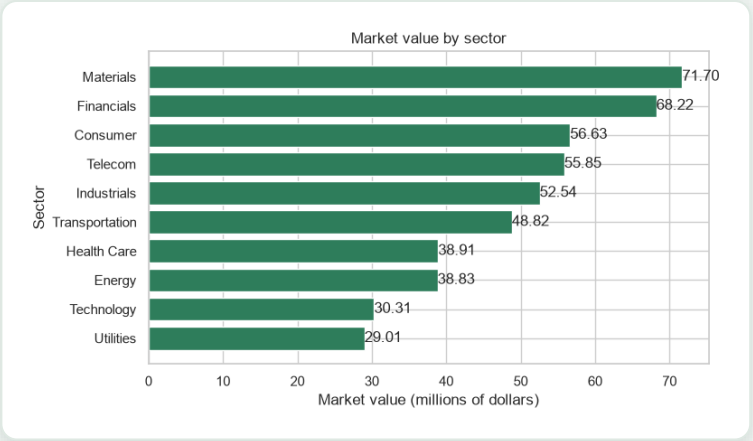
The chart's default, Exclusive median, matches "QUARTILE.EXC"; choose Inclusive median to match pandas.

Count the bonds behind a box, and draw its histogram before trusting it. Now one number per group.

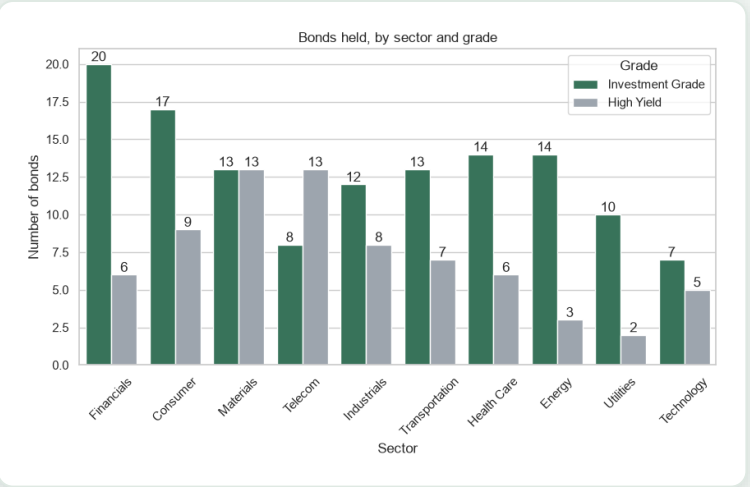
11.5 Bars by bucket, by sector, and by grade



A bar of averages hides the bonds: B's 503 stands against its box's median of 409.5, lifted by the bond at 1,446.



Smallest first puts the largest on top: Materials, then Financials.



Materials alone splits evenly, 13 and 13.

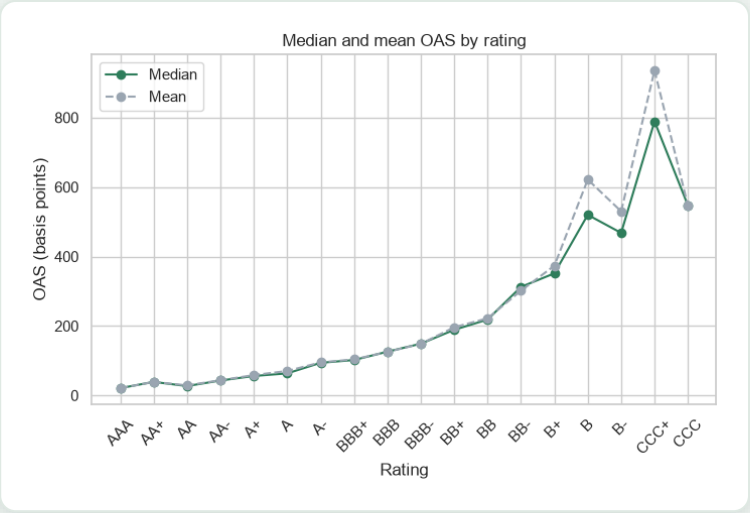
Excel side by side. The bucket is column U.

Job	Python	Excel
One group's average	`the BBB bar, 128`	=AVERAGEIFS(M:M, U:U, "BBB"), 127.82
Upright or flat bars	`ax.bar(), ax.barh()`	Clustered Column, Clustered Bar
Tilted labels	`ax.tick_params(axis='x', rotation=45)`	Format Axis, Custom angle

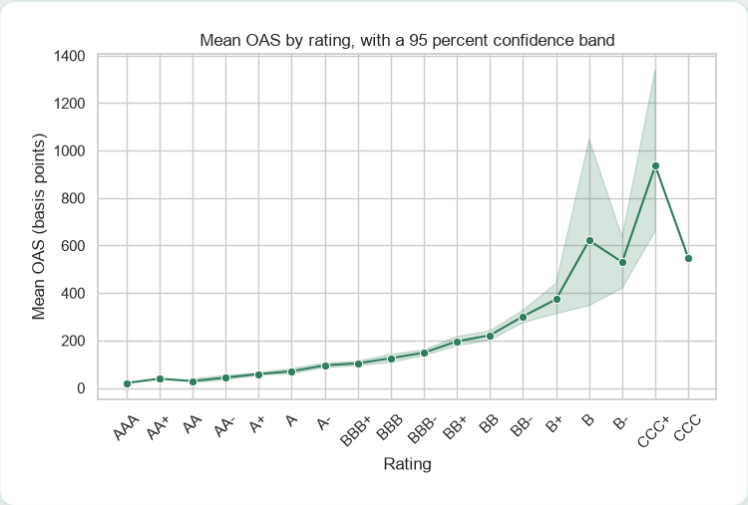
A PivotTable sorts buckets as text, A before AA, so drag them into credit order.

Bars compare named groups. When the groups have an order, a line can join them.

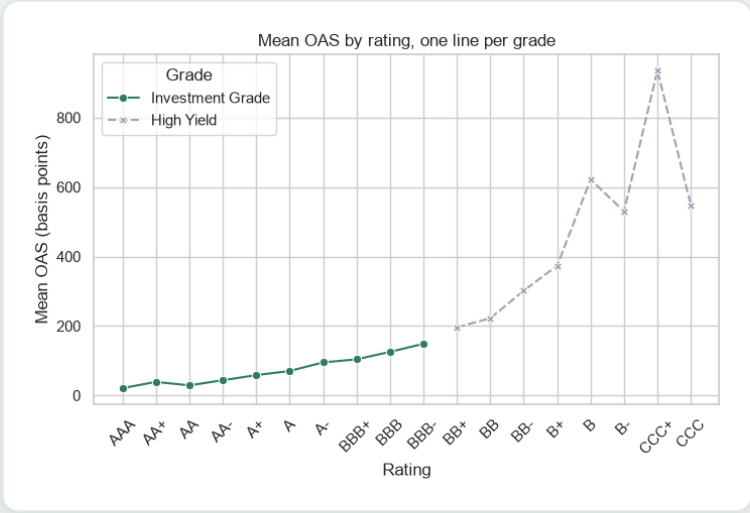
11.6 Lines along the ratings, and a band



The median climbs slowly to 149 bp at BBB-, then steeply, dipping at AA, B-, and CCC.



The band is narrow on the left, wide at B and CCC+, and absent on the single bonds of AA+ and CCC.



The lines never join, because no rating is in both grades.

Excel side by side. The score is column T.

Job	Python	Excel
One group's median	`by_score.loc[10, 'median'], 149`	=MEDIAN(IF(T2:T201=10, M2:M201)), Ctrl+Shift+Enter before dynamic arrays
Line plot	`ax.plot(x, y, marker='o')`	Line with Markers

Dots for the data, a band for how sure the average is. Now two measures at once.

11.7 Duration against OAS, by grade

```
ig = book[book['grade'] == 'Investment Grade']
hy = book[book['grade'] == 'High Yield']
print('Book:', round(book['duration'].corr(book['oas']), 2))
print('Investment Grade:', round(ig['duration'].corr(ig['oas']), 2))
print('High Yield:', round(hy['duration'].corr(hy['oas']), 2))
Book: -0.11
Investment Grade: 0.24
High Yield: -0.1
```

Investment Grade is a flat band under 205 bp, with High Yield mostly above it.

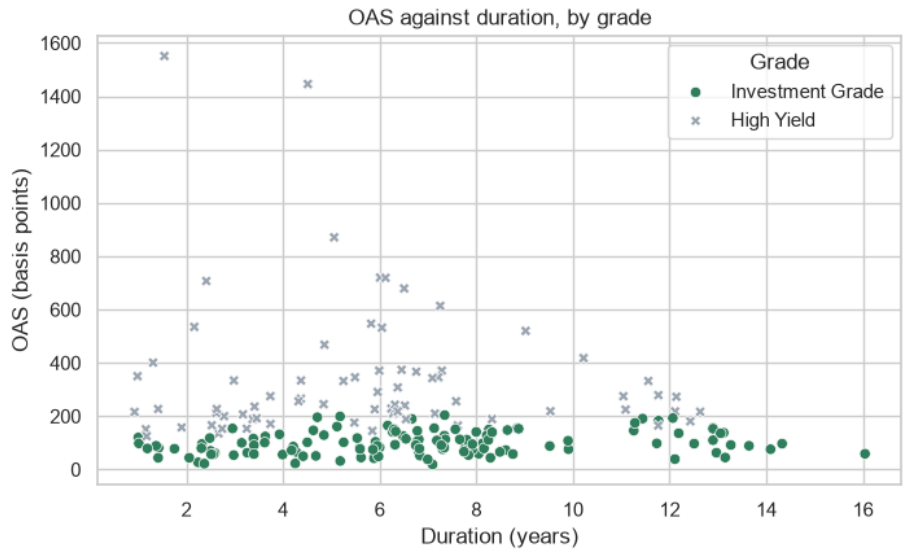
The book's -0.11 describes neither grade, and no correlation says one column causes the other.

In 1973 F. J. Anscombe published four small data sets with the same means, variances, and correlation. Drawn, they look nothing alike.

Source: The American Statistician, 1973, https://doi.org/10.1080/00031305.1973.10478966

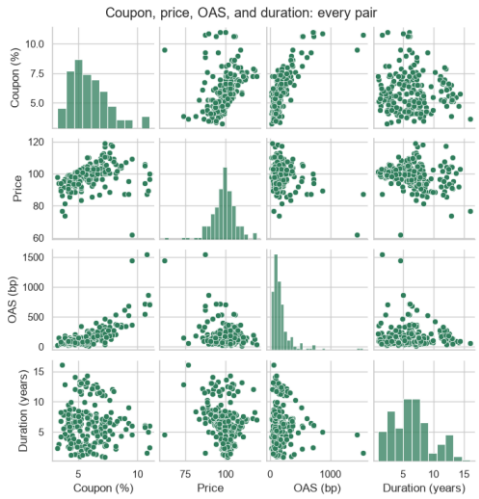
Excel side by side

Job	Python	Excel
Scatter	`ax.scatter(book['oas'], book['yield_pct'])`	Insert, Scatter;
		the first column goes on x



One number for the whole book can describe neither group. Now every pair at once.

11.8 and 11.9 Every pair, then a table in colour

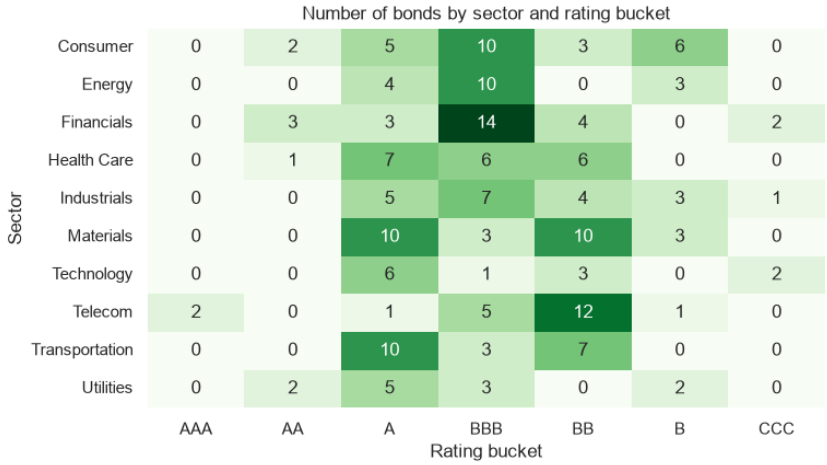


The diagonal holds each column's histogram; every other panel is a scatter, each pair twice, mirrored.

Excel side by side

Excel has no pair plot, so six Scatter charts do its job.

A heatmap is a table with a colour behind every number, as pandas Part 2's "background_gradient".



Financials BBB is the darkest cell, 14 bonds, and both AAA bonds are in Telecom.

The eye finds the dark cells before it reads a figure.

Excel side by side. Duration is column N.

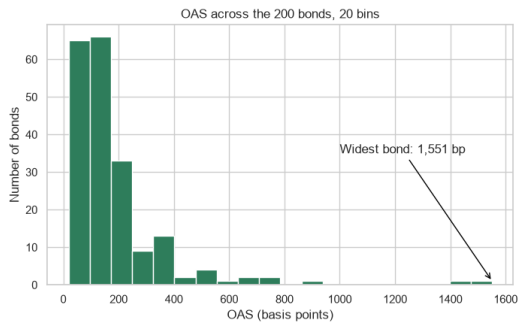
Job	Python	Excel
Correlations	`corr()`	=CORREL(M2:M201, N2:N201);
		for the table, the Analysis ToolPak's Correlation
Colour behind numbers	`sns.heatmap(table, annot=True)`	Conditional Formatting, Color Scales

Predict first: the 5-bin count, the whisker's end

Predict 1

In 5 bins, each bin is four of the 20 above joined, and the first runs from 19 to 325.4 bp. How many bonds land in it? Write down a number.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
counts, edges, bars = ax.hist(book['oas'], bins=5)
ax.bar_label(bars)
ax.set_title('OAS across the 200 bonds, 5 bins')
ax.set_xlabel('OAS (basis points)')
ax.set_ylabel('Number of bonds')
plt.show()
print('Below 250 bp:', (book['oas'] < 250).sum(), ' above 500 bp:', (book['oas'] > 500).sum())
```



Read up from the x axis: the bonds crowd the left, and a thin tail runs out to 1,551.

Predict 2

Where does the upper whisker end? A) At 1,551, the highest value. B) At 409.875, the limit. C) At the widest bond no higher than 409.875.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
ax.boxplot(book['oas'], tick_labels=['All 200 bonds'])
ax.set_title('OAS across the 200 bonds')
ax.set_ylabel('OAS (basis points)')
plt.show()
```

The box spans the quartiles with a line at the median; a whisker reaches toward a limit 1.5 x IQR beyond the box, and bonds past it are drawn alone.

Here the upper limit is 213 + 1.5 x 131.25 = 409.875.

Predict first: seaborn's bar order, yield against OAS

Predict 3

In what order does "sns.barplot(data=book, x='bucket', y='oas')" draw the seven bars? A) Credit order: AAA, AA, A, and on. B) Alphabetical, as "groupby" gives: A, AA, AAA, and on. C) First appearance: BBB, B, BB, and on.

```
fig, ax = plt.subplots(figsize=(8, 4.5))
sns.barplot(data=book, x='bucket', y='oas', errorbar=None, ax=ax)
ax.set_title('Average OAS by rating bucket')
ax.set_xlabel('Rating bucket')
ax.set_ylabel('Average OAS (basis points)')
plt.show()
print(book['bucket'].unique().tolist())
```

The book's first rows are BBB, B, and BB.

Predict 4

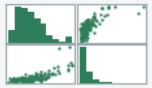
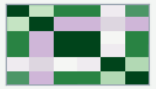
What is the correlation of yield and OAS? A) Near 1. B) Near 0. C) Negative.

```
print('Correlation:', round(book['oas'].corr(book['yield_pct']), 2))
widest = book.loc[book['oas'].idxmax()]
```

A scatter plot places one point per bond by two columns. Correlation measures how close they lie to a straight line: 1 rising, -1 falling, 0 no line.

The note, answered: the question picks the chart

The question picks the chart

The question	Today's chart	Its name	matplotlib, seaborn	Excel
One column's spread		**Histogram**	`ax.hist` `sns.histplot`	Histogram
Groups compared		**Box plot**	`ax.boxplot`	Box and Whisker
One number per group		**Bar chart**	`ax.bar, ax.barh` `sns.barplot, sns.countplot`	Clustered Column or Bar
Change along an order		**Line plot**	`ax.plot` `sns.lineplot`	Line with Markers
Two columns together	Predict 4	**Scatter plot**	`ax.scatter` `sns.scatterplot`	Scatter
Every pair at a glance		**Pair plot**	`sns.pairplot`	a Scatter per pair
A table's large and small cells		**Heatmap**	`sns.heatmap`	Color Scales

Asked	Chart	Answer
Steps by rating	box plot	medians 21 to 707 bp
Where the money sits	bars	Materials 71.70 million

The habits hold for every chart:

Work out the numbers first, and check the chart against them.

Title every chart, and label every axis with its unit.

Know how many bonds stand behind each bar, box, or point.

The question picks the chart.

The next notebook puts a line through a scatter, shows the bonds behind each box, and readies a chart for the meeting.